

Efficient Deepfake Detection via Layer-Frozen Assisted Dual Attention Network for Consumer Imaging Devices

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Abstract—The advancement of open-source frameworks and user-friendly manipulation applications has accelerated the spread of deep fakes. In this study, we proposed optimal features assisted with a dual attention (DA) network strategy to combat this proliferation in consumer imaging devices. We employed EfficientNetV2 (ENV2) as the primary feature extractor, initially utilizing its pre-trained weights from the ImageNet dataset while keeping its layers frozen to leverage their rich feature extraction capabilities. We enhance this base model with a DA module that integrates the Convolutional Block Attention Module (CBAM), which utilizes both channel attention (CA) and spatial attention (SA) mechanisms to improve feature representation. CA dynamically adjusts channel-wise feature responses to capture interdependencies between channels, thereby improving feature discrimination. SA allows the network to focus on important regions within feature maps, enhancing localization and reducing noise. The features are assisted in multi-stages of the network with residual fashion to focus on discriminative visual information. During fine-tuning, we unfreeze the deeper layers of ENV2 for further refinement of learned features to better suit the deepfake dataset. We employed a targeted fine-tuning approach, unfreezing specific layers and applying iterative adjustments to optimize performance, providing valuable insights into countering the growing use of synthetic media in consumer imaging. To validate our network, we conducted comprehensive experiments on four benchmark datasets: FaceForensics++, the World Leaders, Celeb-DF, and DFDC. As a result, our network achieved superior performance compared to existing benchmarks and state-of-the-art approaches, offering a promising solution for robust deepfake detection (DD) in consumer imaging technologies.

Index Terms—Deepfake detection, consumer imaging technologies, attention mechanisms, model optimization, progressive fusion, fine-tuning.

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I. INTRODUCTION

CONSUMER Electronics, especially smartphones and cameras, have transformed media creation and sharing. While inexpensive, high-quality CE has made content creation more accessible [1], it has also contributed to the development of deepfake technology, raising questions about authenticity. Deepfakes pose a widespread hazard, affecting the privacy of public and private people, especially celebrities, politicians, high-security personnel, and other public figures. This spreads false information and scams, compromising personal security, damaging reputations, and undermining public trust in media. Malicious actors exploit these technologies for misinformation, which comes under cyber crime [2]. Hence, face privacy preservation is often considered a burning issue [3]. To tackle this issue, strong deepfake detection (DD) systems are required to be integrated into consumer devices. There is an urgent need to detect fake content to safeguard security and confidence in the use of digital media [4]. CE companies are more interested in addressing challenges posed by deepfake technology to retain their brand reputation and customer trust [5].

The emergence of deepfake technology has resulted in situations where facial features in pictures and videos have been transformed [6], [7]. This has led to a decrease in trust and safety in digital media. Deepfake technology significantly affects consumer appearance as consumers increasingly depend on visual media for information, entertainment, and communication [8]. Therefore, it is crucial, especially in today's digital era, to develop strong networks that are capable of recognizing forgeries to tackle the increasing challenges related to security and trust in consumer imaging technology. Most of the attempted methods currently in use focus on particular varieties of deepfakes. For instance, [9] suggested a technique for spotting lip-syncing deepfakes. Their approach took advantage of the discrepancies between a phoneme (spoken word) and a viseme (mouth form). Another attempt [10] proposed a data augmentation-based method, Face-Cutout. As from the name, they cut the irrelevant facial landmark information and then selected only relevant features and parts of video frames. To extract facial landmark features, the study [11] employed the open-source toolbox OpenFace2. These generated features were combined with an action unit to train a binary SVM for DD. The work [9] proposed the end-to-end system MesoNet to identify phony Face2Face videos and

deep fakes. Moreover, the work [12] used ensemble learning while using four architectures of CNN. They used data augmentation like downsampling, hue saturation, JPEG compression, etc. during training and validation for the robustness of the model. The study proposed a TAR approach [13] with the utilization of a transfer learning-based autoencoder with Residuals to detect several types of deepfakes. It is observable that autoencoders are extensively deployed in many domains with good results [14]. As the study used FF++ to train the model and evaluated it on real-world videos of almost 200 counts containing 50 different celebrities, it became evident that there is a continuous need for intelligent advancements in the field. These advancements, trained on large amounts of data with the progressive integration of attention modules and effective fine-tuning, are crucial for enhancing security and confidence in consuming digital content.

A. Limitations of the Related Literature

The limitations of existing studies in the domain highlight the need for feasible techniques for deployment on consumer devices. One of the significant limitations of most existing methods is the inadequate capacity to capture complex features precisely. This limitation is often based on the reliance on the direct utilization of intermediate feature extractors without proper empirical validation at the early stages of the networks. Moreover, the lack of optimal features is unable to be learned with shallow layered networks to grasp the important patterns before the ultimate prediction. Attention-based techniques have shown promise in enhancing feature representation and discriminability. However, their utilization in existing architectures is often technically unsystematic, with less representative features. As a result, the learned features may not be effectively refined and optimized. This can result in less discriminative and less informative feature representations, hindering the model's ability to detect DD accurately. Moreover, the lack of generalized optimization methods for the utilized modules can impact the overall effectiveness of the network, especially when faced with evolving synthetic media manipulation methods. These limitations emphasize the immediate need for feasible solutions that directly tackle the challenges mentioned above, thereby enabling the design of a more effective network.

B. Offered Contributions

- We introduced a new network that adeptly captures the complex features of both fake and real faces. Our novel and progressive fusion strategy significantly boost performance, making it feasible to widely implement DD into consumer devices such as smartphones to quickly detect and alert authorities about potentially suspicious content before sharing it on social media platforms.
- Optimal intermediate features are extracted with frozen initial layers of ENV2 in the initial training phase to utilize pre-trained weights effectively. Integrating a DA strategy with an optimized dilated convolutions block enhances feature extraction. The CBAM refines features by focusing on critical patterns and reducing noise. At

the same time, the optimized dilated convolutions block captures both local details and global context effectively, expanding the receptive field without significantly increasing computational costs.

- The proposed fine-tuning strategy includes training the model initially for 20 epochs and then fine-tuning the more profound layers while keeping the first 1000 layers of ENV2 frozen. This selective layer training enables the model to adapt to the specific features of the large datasets while maintaining stability and preventing overfitting. The fine-tuning process, which extends from epoch 21 to 30, results in a significant accuracy improvement.
- Our progressive modifications underwent an extensive assessment using four benchmark datasets: FF++, WLDR, CELEB-DF, and DFDC. The comprehensive evaluations demonstrated that our contributions outperformed state-of-the-art (SOTA) networks, highlighting the significance of our network as the best choice. We conducted extensive intra-class cross-experiments to assess the generalization capabilities of the proposed network. Our proposed network consistently maintains its position as the benchmark in safeguarding consumer trust and privacy in a rapidly evolving digital environment.

The remaining manuscript is organized: Section II reviews the detailed literature review, Section III discusses our proposed methodology, Section IV details experimental work and results discussion, and our job is concluded in Section V.

II. RELATED LITERATURE

Various approaches have been proposed to tackle the DD challenge, leveraging both traditional and advanced techniques in machine learning (ML) and deep learning (DL) [15]. To address the unique demands of DD, researchers have explored many methodologies for DD [16]. For instance, the work [11] used videos of individuals to extract facial expressions and head movements from hours-long videos. They further extracted features from a 10-second clip, as frame by frame and 190-D features were found by them. SVM was used to classify the 10-second clip as being consistent or not with the learned mannerisms of an individual. The attempted technique requires much effort to propose a model for each, and hand-crafted features might not be optimal, highlighting the complexity inherent in securing CE against deepfake threats. Another framework called FakeCatcher was proposed [17] for DD using signals from three regions of the face in actual videos. The method presented a promising way for DD, especially in CE devices, where user interaction is vital. Moreover, the study [18] proposed a method that used CNN to extract features from video frames and then feed them to LSTM to predict sequential patterns. However, the technique was evaluated on randomly collected videos from different websites rather than from a single standard dataset, which may limit the test of the generalization.

The study [19] proposed a technique, Face X-ray-based forgery detection, which used a greyscale image and used to find whether this image can be decomposed into the blending of two images from various sources. Another method [9]

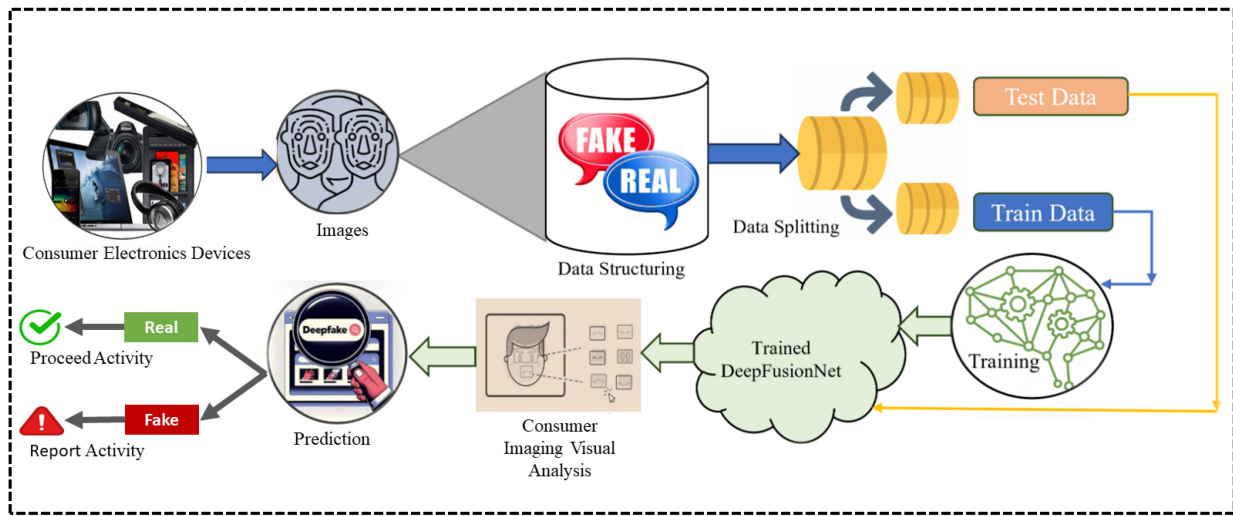


Fig. 1. Conceptual Overview of the Proposed Network Suggested for Deployment on Consumer Devices.

proposed Mesonet to detect deepfakes with an end-to-end architecture with stacked layered CNNs. The work [20] proposed a vision transformer (ViT)-based model which used distillation techniques, along with pre-trained EfficientNetB7. They trained the network on DFDC by passing the patches extracted from EfficientNet combined with ViT. These combined patches are then passed to the transformer encoder. By considering the previous limitations of DD methods, the study [10] proposed a data augmentation-based method, Face-Cutout. They cut the irrelevant facial landmark information and selected only relevant features and parts of video frames. The study [21] developed a DefakeHop technique that uses successive space learning for auto-feature extraction from different regions of the forged image. The features are extracted by c/w Saab Transform and processed through feature distillation using spatial dimensionality reduction and soft classification for a concise description of images. Furthermore, the work [13] a transfer learning-based autoencoder with Residuals to detect several types of deepfakes. They achieved 89.49% zero-shot accuracy on this evaluation. Additionally, the study [22] used CNN and ViT to classify a video as Real or fake. They used CNN for learned feature extraction while these learned features were passed as input to the ultimate network for predictions. A capsule network was suggested in the study [23] for exposing various kinds of images and video fake moments.

In addition, the work [12] used ensemble learning while using four architectures of CNN with attention. The research [24] used GANs-based models, called MRI-GAN for DD. Their technique consists of perceptual differences between images to detect forged videos and a plain-frames-based model. DFDC was used to train the model achieving 91% test accuracy for the plain-based model and the MRI-GAN model achieved 74% for perceptual differences in images. The research [25] proposed a unified multimodal framework that extracts learned channels from audio and visual modalities, mixes them independently and then processes them jointly before feeding the combined representation to a multilabel classification head. The study [26]

also used audio and visual features of a video to train the model on multi-modality features. Their work uses the Siamese approach to learn audio and visual features side by side. The authors used DFDC and Deepfake-TMIT datasets for the evaluation of their proposed framework. Another study [27] tackled the DD problem by utilizing spatiotemporal properties in the video frames. Additionally, this design works well on the substantial face-swap dataset DFDC, but it is highly costly and computationally expensive to propose for deployment. The work [28] proposed a convolutional brain organization to extricate spatial and transient irregularities in deepfakes made with GANs. GANs-created deepfakes are outwardly convincing nevertheless, they have a few spatial and transient relics that can be taken advantage of for identification errands. The researchers proposed the above-mentioned approaches for DD. However, there is still major room for developing optimal features-assisted visual intelligence techniques to effectively grasp the complex features of deepfakes.

III. PROPOSED METHODOLOGY

The proposed methodology emphasizes extracting deep features to enhance the robustness of intermediate representations. We investigate effective modifications in CBAM to refine these features. Additionally, we introduced an optimized convolution block with dilated convolution layers and GAP. Furthermore, we analyze the impact of fine-tuning and incremental progressive training on overall performance. As illustrated in Figure 2, our proposed network integrates these components to enhance accuracy and robustness, specifically designed for CE applications.

A. Deep Features Extraction

Our proposed technique incorporates the ENV2 model as the primary feature extractor. The model is filled with pre-trained weights from the ImageNet dataset and set to be non-trainable, meaning the pre-trained weights are used without being updated during training. This enables the ultimate network to take advantage of rich feature extraction ability

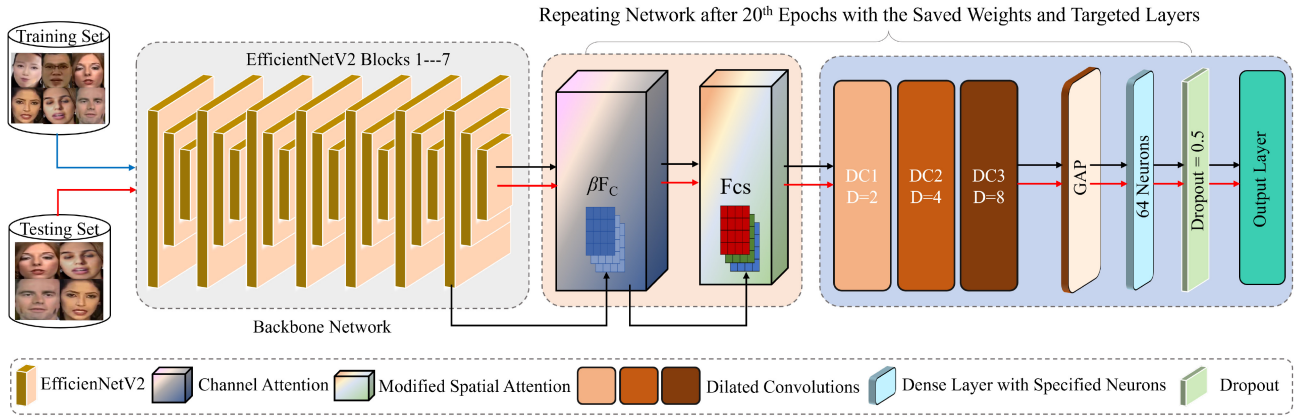


Fig. 2. Visual Overflow of the Integrated Modules in the Proposed DD Network.

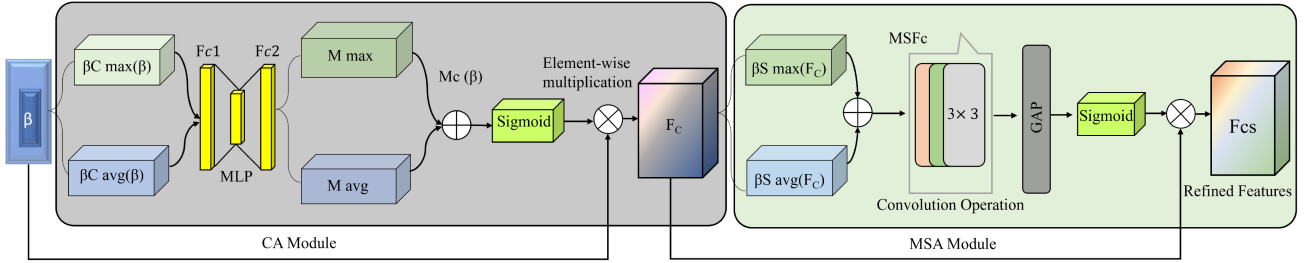


Fig. 3. CBAM Module with CA and SA for Feature Refinement.

without making any modifications to the pre-trained weights. In order to further optimize the model, we incorporate several progressive components with DA. The CBAM improves features by iteratively computing attention mappings along the channel and spatial dimensions. This enhances the capacity to identify features with feasible skip connections to assist the mature features with initial-level features. In addition, the block dynamically adjusts channel-wise feature responses by explicitly capturing the relationships between channels, hence boosting the network's ability to enhance information. The combination of these progressive improvements, together with GAP forms a strong feature extraction architecture. The model runs an initial training process for 20 epochs, during which it achieves an accuracy as mentioned in Table III. The resultant weights are then saved for fine-tuning usage.

B. Convolutional Block Attention Module

Attention mechanisms are extensively explored in many domains with excellent performances [29], [30], [31], [32], [33], [34]. Taking this motivation, we integrated CBAM, which effectively boosts the network's ability to capture and refine features at multiple stages. CBAM receives the intermediate features β for dynamic recalibration of CA and SA, which progressively improves feature representation. This utilization reduces noise information while focusing only on channels and spatial regions. It significantly enhances the potential of the network to focus on various types of critical and complex deepfakes, as highlighted in the qualitative results. The CBAM architecture contains a dual attention scheme where the CA module controls inter-channel functions of

features, considering each channel's feature map. As shown in Figure 3, CBAM architecture with CA and SA includes a GAP layer, a maximum pooling layer, fully connected layers, and a convolution operation. During training, convolution feature maps of each channel do not equally represent an object. Therefore, we incorporate both maximum and average pooling strategies to obtain separate spatial background descriptions, β_{Cavg} and β_{Cmax} , through average and maximum pooling operations on the spatial dimension of the feature map β , respectively.

$$\beta_{Cavg} = \text{Avg-Pool}(\beta) \quad (1)$$

$$\beta_{Cmax} = \text{Max-Pool}(\beta) \quad (2)$$

Two FC layers, f_{c1} and f_{c2} , process β_{Cavg} and β_{Cmax} , respectively, followed by ReLU activation to obtain M_{max} and M_{avg} . These are then added to produce the weight coefficients $M_c(\beta)$, maintaining feature flow through a skip connection:

$$M_{avg} = \sigma(f_{c2}(\text{ReLU}(f_{c1}(\beta_{Cavg})))) \quad (3)$$

$$M_{max} = \sigma(f_{c2}(\text{ReLU}(f_{c1}(\beta_{Cmax})))) \quad (4)$$

$$M_c(\beta) = M_{avg} \oplus M_{max} \quad (5)$$

The CA map F_c is obtained by applying an addition function to M_{max} and M_{avg} , facilitating feature refinement:

$$F_c = M_c(\beta) \otimes \beta \quad (6)$$

The SA module complements the CA module by focusing on inter-spatial relationships and identifying the most important regions. Apart from the benchmark way [35], we reduced the number of kernel sizes from 7×7 to 3×3 . This reduction

aims to enhance computational efficiency and reduce model complexity while still maintaining effective feature refinement through the CBAM mechanism. It creates two-dimensional feature maps, β_{Savg} and β_{Smax} , using average and maximum pooling operations on the channel dimension of F_c . These feature maps are then combined and convolved to produce a refined two-dimensional SA feature map, employing convolutional layers for refinement.

$$\beta_{Savg} = \text{Avg-Pool}(F_c) \quad (7)$$

$$\beta_{Smax} = \text{Max-Pool}(F_c) \quad (8)$$

After applying the GAP on the feature maps, we obtain the SA map Ms_{F_cGAP} , which is then concatenated with F_c to create the refined feature map F_{cs} . This feature map is passed to the customized ConvNet module before ultimate prediction.

$$Ms_{F_c} = \sigma(f^{3 \times 3}(\beta_{Savg} \oplus \beta_{Smax})) \quad (9)$$

$$Ms_{F_cGAP} = \text{GAP}(Ms_{F_c}) \quad (10)$$

$$F_{cs} = Ms_{F_cGAP} \otimes F_c \quad (11)$$

C. Optimized Dilated Convolutions Block

The customized dilated convolutions block receives the features from CBAM and enhances features by leveraging dilated convolution layers with varying kernel sizes. Initialized with the input features, the block sequentially applies 3x3 convolutions with dilation rates of 2, 4, and 8, respectively, each followed by ReLU activation and padding to maintain spatial dimensions. This strategic configuration enables the convolutional filters to expand their receptive fields exponentially without significantly increasing computational costs. As a result, the network is able to capture both local details and global context more effectively, which is crucial for tasks requiring fine-grained feature analysis and semantic understanding. Subsequently, the feature maps undergo GAP to distill spatial information into compact representations, facilitating dimensionality reduction while retaining essential features. Additional dense layers with ReLU activation refine feature representations, promoting nonlinear transformations that enhance discriminative power in the extracted features. The features are then passed to the classifier layers for ultimate predictions.

D. Fine Tuning

Following the initial training phase till the 20th epoch, the model undergoes fine-tuning to optimize its performance further for the next 10 epochs. The ENV2 model fine-tunes its deeper layers by using the first 1000 layers as a fixed feature extractor with frozen weights. The remaining 329 layers are set to be trainable, allowing adjustments to better suit our dataset. This strategy allows the model to refine learned features to better fit the specific weights of our large datasets. Throughout the fine-tuning phase, the model retains CBAM, ConvNet with dilated convolutions, GAP, dense layers, and dropout layers from the original setup. Fine-tuning continues beyond the initial 20 epochs, spanning from epoch 21 to 30. This phase allows for adjustments in the weights of the deeper layers to better fit our dataset while leveraging the

TABLE I
SUMMARY OF DATASETS

Dataset	Original Videos	Fake Videos
FF++	1,000	5,000
WLDR	-	-
Celeb-DF	590	5,639
DFDC	23,564	104,500

stability and generalization benefits provided by the frozen layers. The work [36] also practiced training plasticity to selectively freeze well-trained layers in deep neural networks. Our attempted fine-tuning process substantially enhances the accuracy, improving from 94% to 98% on FF++. This highlights the effectiveness of our approach, combining pre-trained knowledge with dataset-specific parameter refinement to achieve optimal performance.

IV. RESULTS AND DISCUSSION

This section includes the experimental configuration, the dataset used, and the outcomes of assessing the proposed network. The ablation study and empirical analysis of the integration of different modules are elaborated. It also includes a comparison with existing methods and the results of several intermediary models.

A. Experimental Setup

We implemented our proposed network in the TensorFlow DL framework using an NVIDIA GeForce RTX 3090 GPU. We employed an Adam optimizer with a momentum of 0.9 and set the initial learning rate to 1×10^{-4} . The network was initially trained for 20 epochs with the base model frozen. During the final 10 epochs of the fine-tuning stage, the learning rate was reduced to accommodate the larger model and to readapt the pre-trained weights, thereby avoiding overfitting. The learning rate was further reduced by a factor of 0.5 whenever the validation accuracy plateaued for two consecutive epochs. We used the ReduceLROnPlateau callback to adjust the learning rate based on validation performance. Training images were resized to 160×160 and the batch size was set to 32 as empirically validated. We employ common metrics to evaluate our method, including Area Under the Receiver Operating Characteristic Curve (AUC), Equal Error Rate (EER), Accuracy (ACC), Precision, Recall, and F1-Score. The AUC value ranges from 0 to 1, where a value closer to 1 indicates a better performance. The EER is the point on the ROC curve where the false acceptance rate equals the false rejection rate.

B. Datasets

We used four benchmark datasets to conduct the empirical analysis to evaluate the proposed network. Table I shows the summary of the utilized datasets in our study. The datasets include FF++ [37], Celeb-DF [38], WLDR [39], and DFDC [40]. FF++ is a popular dataset that includes videos of five commonly used facial manipulation techniques: Deepfakes (DF), Face2Face (F2F), NeuralTextures (NT),

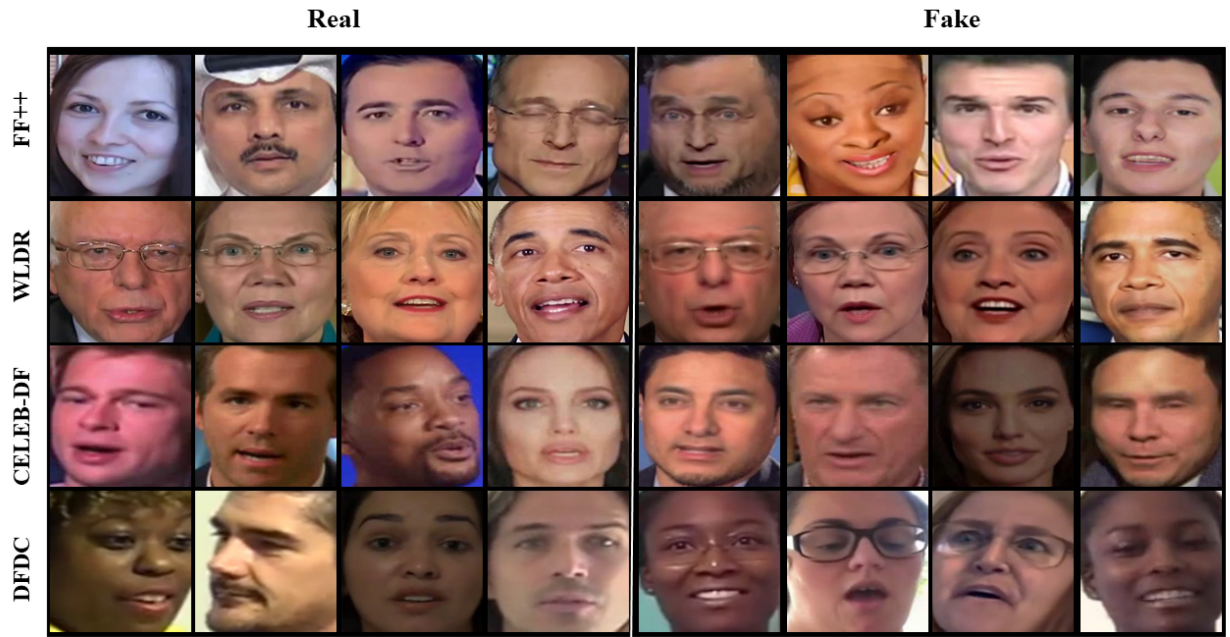


Fig. 4. Visual illustrations of the fake and real samples taken from the utilized four datasets.

FaceSwap (FS), and FaceShifter (FST). The majority of the frames in the dataset show the frontal face. For each class, we utilized approximately 72,000 frames. The **WLDR Dataset** comprises videos featuring prominent politicians and presidents such as Obama and Clinton. The video clips show politicians speaking in continuous dialogue, with every clip focusing on a single individual. This dataset includes several manipulation techniques, such as face-swapping, lip-syncing, and impersonation. We extracted almost 38,715 fake frames and 163,177 real frames. The dataset **Celeb-DF** has videos sourced from YouTube, along with fake videos of high quality. This dataset includes many celebrity interview clips that are of good quality and have consistent background information. We worked with 45,000 real and 45,000 fake frames from it. The dataset **DFDC** is very challenging due to the diverse manipulation methods. Our study included 89,572 fake frames and 26,074 real frames.

C. Quantitative Analysis

1) *Intermediate Features Analysis*: Table II presents a comprehensive analysis of various backbone models with the proposed network on the FF++ dataset to assess intermediate-level features. The reported performance metrics, including accuracy, precision, recall, and F1 score, highlight our proposed network's effectiveness compared to several architectures. MobileNetV2 achieved an accuracy of 89.50%, with precision at 85.04%, recall at 88.01%, and an F1 score of 86.53%, indicating reasonably good performance but still leaving room for improvement. VGG16, ResNet, and InceptionV3 models demonstrated comparatively good feature representation capabilities, benefiting from their deeper architectures. DenseNet and EfficientNet series showed progressive improvements with increasing model complexity, highlighting their scalability and efficiency. In a nutshell, the tabulated

TABLE II
INTERMEDIATE FEATURES ANALYSIS

Model	Accuracy (%)	Precision	Recall	F1 Score
MobileNetV2	89.50	85.04	88.01	86.53
VGG16	90.22	88.01	89.52	88.73
ResNet50	91.45	90.54	91.05	90.76
ResNet101	91.84	91.23	91.55	91.32
InceptionV3	92.27	91.87	92.02	91.92
DenseNet121	92.34	91.53	92.50	92.01
DenseNet169	93.01	92.83	93.20	93.01
DenseNet201	93.26	93.00	93.53	93.25
EfficientNetB0	92.55	91.74	92.31	92.01
EfficientNetB1	93.29	92.58	93.03	92.73
EfficientNetB2	93.67	93.25	93.53	93.30
EfficientNetB3	94.01	93.85	94.01	93.93
EfficientNetB4	95.46	95.02	95.24	95.13
EfficientNetB5	95.89	95.56	95.73	95.62
EfficientNetB6	96.19	96.01	96.00	96.01
EfficientNetB7	97.56	97.34	97.45	97.35
DFNet (Ours)	98.54	98.48	98.59	98.54

performances demonstrate that the proposed network surpasses existing models across all key performance metrics, confirming the efficacy of the proposed enhancements in feature extraction and model architecture, making it a highly reliable choice. Our proposed network significantly outperformed all other models, achieving an outstanding accuracy of 98.54%, with precision at 98.48%, recall at 98.59%, and an F1 score of 98.54%. These results indicate that the fine-tuning of progressive feature extraction, CBAM, and customized dilated convolution block effectively captures and refines features for robust DD.

2) *Progressive Module Integration*: Table III presents the analysis of various module integrations to observe the impact of each block progressively. Starting with the ENV2, which achieved an accuracy of 84.53% on FF++ and 83.55% on WLDR. Adding a CA module to ENV2 enhanced the

TABLE III
PERFORMANCE ANALYSIS ON FF++ AND WLDR WITH INTEGRATED MODULES

Dataset	Model	ACC	Precision	Recall	F1 Score	AUC
FF++	ENV2	84.53	84.05	85.26	84.65	92.84
	ENV2 + CA	87.25	86.70	87.10	86.80	93.12
	ENV2 + CA + SA	88.60	87.95	88.45	88.30	94.75
	ENV2 + CBAM RES	90.44	90.00	90.99	90.49	96.85
	ENV2 + CBAM RES + ConvBlock	94.34	94.30	94.40	94.35	98.83
	Proposed (ENV2 + MCBAM + ConvBlock + Fine Tuning)	98.54	98.48	98.59	98.54	99.83
WLDR Dataset	ENV2	83.55	82.95	83.40	83.10	88.75
	ENV2 + CA	85.20	84.50	85.10	84.85	90.45
	ENV2 + CA + SA	86.65	86.10	86.55	86.20	91.90
	ENV2 + CBAM RES	88.30	87.70	88.25	88.00	93.80
	ENV2 + CBAM RES + ConvBlock	92.95	92.55	92.85	92.75	96.65
	Proposed (ENV2 + MCBAM + ConvBlock + Fine Tuning)	96.75	97.20	96.85	97.10	98.07

TABLE IV
CROSS-DATABASE EVALUATION FROM FF++ TO CELEB-DF AND DFDC IN TERMS OF AUC AND EER. THE FF++ BELONGS TO THE INTRA-TESTING RESULTS, WHILE OTHERS REPRESENT THE UNSEEN DATASET TESTING

Method	FF++		Celeb-DF		DFDC	
	AUC (%)	EER (%)	AUC (%)	EER (%)	AUC (%)	EER (%)
GFF [41]	98.4	3.8	75.7	32.2	71.7	34.6
Local-relation [42]	99.2	3.0	78.3	29.1	76.6	32.2
Face X-ray [43]	87.5	4.2	75.7	26.5	70.1	31.4
DCL [44]	99.3	3.2	82.3	26.4	76.8	31.8
EN-b4 [45]	99.2	3.3	68.6	35.3	70.2	34.4
MAT(EN-b4) [46]	99.3	3.3	76.6	32.5	67.9	38.2
UIA-ViT [47]	99.3	3.2	82.5	27.4	75.9	30.0
F3-Net [48]	98.1	3.5	71.2	33.7	72.9	33.2
MLDG [49]	98.6	3.4	74.5	30.81	71.9	34.4
Xception [50]	99.1	3.7	65.3	38.7	70.0	35.3
LTW [51]	99.2	3.3	77.2	29.3	74.6	33.6
DFNet (Ours)	99.8	2.9	87.8	22.8	89.2	21.7

performance to 87.25% on FF++ and 85.20% on WLDR, with corresponding increases in other matrices. Further integrating SA with CA resulted in an accuracy of 88.60% on FF++ and 86.65% on WLDR, showcasing the additional assistance of combining both attention mechanisms. The utilization of the CA and SA with the practiced residuals (CBAM RES) markedly improved the result to 90.44% on FF++ and 88.30% on WLDR, along with a substantial increase in other performance metrics. It is observable that incorporating a customized convolution block with CBAM RES boosts the accuracy to 94.34% on FF++ and 92.95% on WLDR, demonstrating the effectiveness of refining features. The proposed network, which includes optimal features assisted by modified CBAM (MCBAM), the convolution block, and fine-tuning, achieved the best outcomes with a performance of 98.54% on FF++ and 96.75% on WLDR. The results clearly indicate that the fine-tuning with progressive integrations of the utilized modules.

3) *Intra Class Data Validation*: Table V provides a detailed cross-evaluation of several prominent evaluation methods to assess generalization across data divisions of DF, F2F, FS, NT, and FST. These divisions encompass a wide range of techniques and scenarios, making them a robust testbed for evaluating detection methodologies. Among the performances of the tabulated methods, our proposed network consistently emerged as the frontrunner. The reported trends in the results highlight the robustness of detecting complex manipulations. It is important to highlight that when the

TABLE V
CROSS-DATA EVALUATION ON DIFFERENT DATA DIVISIONS TO ASSESS GENERALIZATION ABILITY OF THE PROPOSED NETWORK

Training Set	Methods	DF	F2F	FS	NT	FST
DF	Xception [50]	99.41	72.66	45.90	81.73	56.81
	FTCN [52]	99.27	79.83	50.71	86.34	52.28
	LipForensics [41]	99.64	73.81	43.6	90.69	47.94
	GFF [53]	99.61	76.4	49.7	80.44	51.93
	Face X-ray [43]	98.9	67.42	75.10	91.8	77.10
	DFNet (Ours)	99.72	75.93	78.62	93.51	77.84
F2F	Xception [50]	81.34	99.3	76.6	65.85	54.29
	FTCN [52]	82.99	99.40	78.42	83.17	58.91
	LipForensics [41]	91.40	99.11	74.1	92.3	55.78
	GFF [53]	84.68	99.33	98.7	98.25	57.31
	Face X-ray [43]	93.61	98.4	83.8	91.52	61.32
	DFNet (Ours)	89.77	99.63	93.91	99.14	62.47
FS	Xception [50]	71.22	69.03	99.4	73.5	60.71
	FTCN [52]	90.16	79.7	99.10	84.1	57.47
	LipForensics [41]	60.15	68.6	98.52	64.9	62.29
	GFF [53]	70.51	99.14	99.5	98.0	62.32
	Face X-ray [43]	86.8	92.19	98.1	89.7	66.81
	DFNet (Ours)	87.89	99.35	99.81	97.90	72.42
NT	Xception [50]	84.5	79.4	57.6	99.48	62.15
	FTCN [52]	88.78	84.6	64.9	98.9	58.72
	LipForensics [41]	94.39	89.77	61.4	97.1	56.45
	GFF [53]	88.41	99.43	99.32	99.11	60.38
	Face X-ray [43]	90.57	97.7	78.85	98.79	59.12
	DFNet (Ours)	96.18	98.29	99.41	99.83	64.27
FST	Xception [50]	74.21	75.12	78.33	79.40	99.25
	FTCN [52]	75.61	77.83	80.47	81.11	99.40
	LipForensics [41]	73.45	74.52	77.29	78.38	99.32
	GFF [53]	63.48	71.31	58.28	72.12	99.36
	Face X-ray [43]	76.82	78.29	80.13	81.78	99.27
	DFNet (Ours)	79.15	80.42	82.35	83.25	99.41

network is trained over the DF division, it achieves the highest accuracy of 99.72% on DF and 93.51% on NT. The proposed network also showed optimal results when trained on F2F,

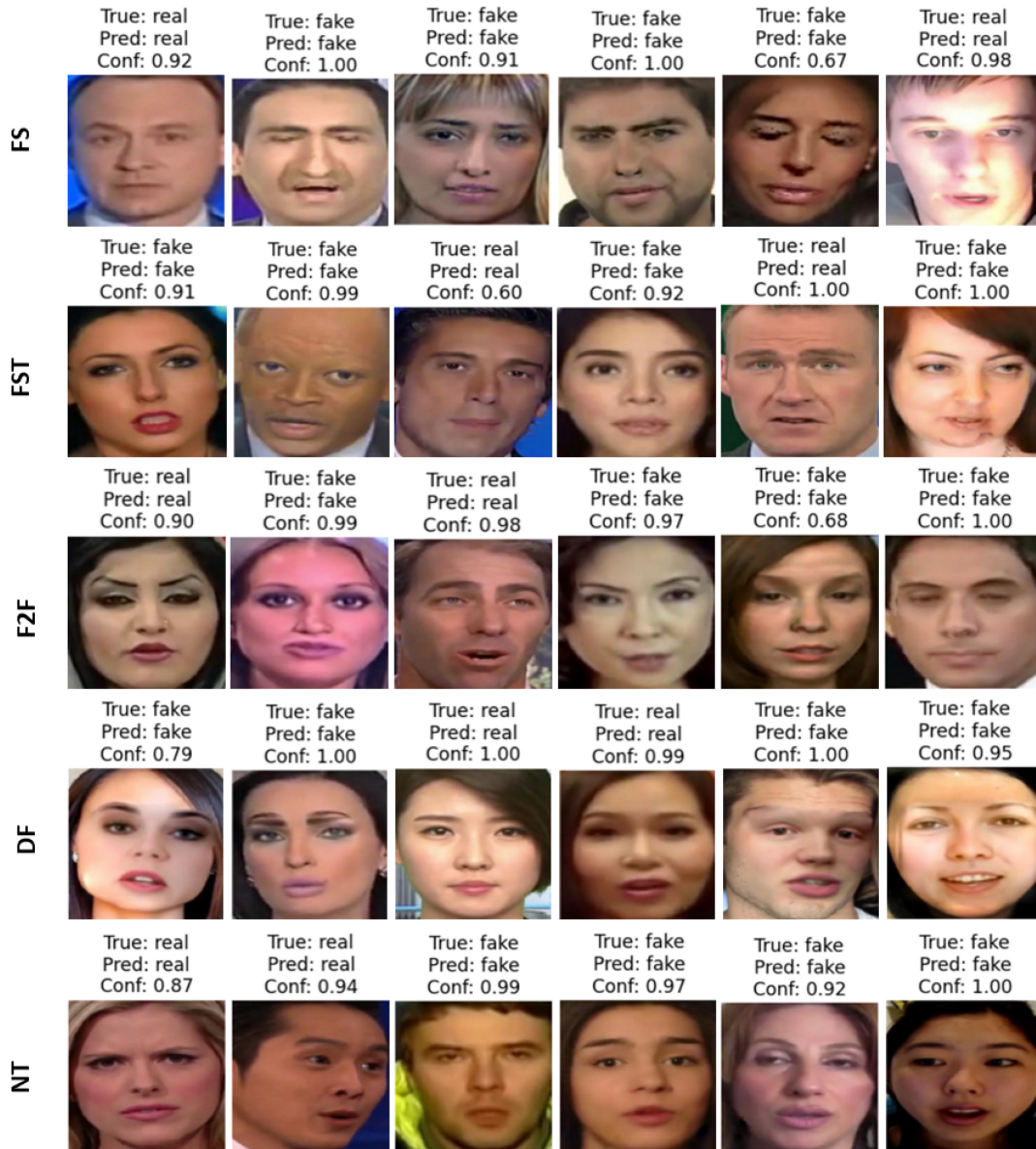


Fig. 5. Qualitative Analysis of Network Predictions Across Different Deepfake Manipulation Classes.

FS, NT, and FST data divisions, outperforming other methods like Xception, FTCN, and Face X-ray. These reported performances highlight that the proposed DFNet is effective and robust in detecting deepfakes across various data evaluation splits. The generalization capability in our context defines the capacity to effectively detect deepfakes across various divisions that vary in manipulation, and quality alteration approaches. Furthermore, the superior performance of the network stems from its ability to capture complex patterns specific to deepfake manipulations. Our network demonstrated effectiveness across different data divisions positioning it as a promising solution in the domain.

D. Comparative Analysis

Table IV highlights the performance metrics of SOTA DD methods across FF++, Celeb-DF, and DFDC datasets,

using AUC and EER as evaluation criteria. It is observable that higher AUC values indicate superior discrimination performance, while lower EER values denote better overall accuracy. The reported performances highlight that the proposed approach consistently surpasses other methods across all datasets, underscoring its efficacy in detecting diverse deepfake manipulations. The performance superiority emphasizes that the network's training strategy, which integrates advanced feature extraction capabilities, optimal DA, customized modules, and robustness enhancements, is well-tailored for tackling the complexities of DD. The significant performance gap observed between the proposed method and existing approaches further validates its superior generalization capability. These findings position the proposed network as a leading solution in the evolving landscape of DD, offering technical insight into its strengths and contributions within the field.

E. Qualitative Results

Figure 5 represents the visual predictions of the proposed network on the FF++ dataset across different deepfake manipulation classes, including FS, FST, F2F, DF, and NT. Each row in the figure corresponds to a specific manipulation technique, with columns representing individual sample predictions. For each sample, the ground truth label, predicted label, and confidence score are provided. The first row exhibits results for FS modifications. The network demonstrates an impressive level of accuracy in differentiating between real and fake images, accompanied by excellent confidence scores. All predictions for the fake class are found true, indicating strong detection ability against FS alterations. The second row represents the outcomes of FST which consistently shows outstanding accuracy by often correctly predicting with a high level of confidence. Despite detecting one real image with a reduced confidence of 0.60, the total average score remains robust. The visual overview of F2F variations is placed in the third row where the network shows excellent detection capabilities, correctly distinguishing the complex features. The fourth row presents the outcomes of DF modifications where the network effectively categorizes most of the images, with confidence values ranging from 0.79 to 1.00. The last row indicates the results for NT modifications. From the visual samples, it can be observed that the network consistently achieves excellent accuracy by accurately predicting with average confidence scores above 0.94 for six given samples. Consequently, the qualitative outcomes presented in Figure 5 indicate that the suggested approach achieves an outstanding level of confidence when detecting multiple types of deepfake variations.

V. CONCLUSION

In this study, we introduced an innovative strategy utilizing optimal features enhanced by a DA mechanism, incorporating the CBAM to dynamically adjust channel-wise feature responses and focus on important spatial regions. The intermediate features are assisted and strengthened by CBAM and customized dilated convolution block. By fine-tuning our network with progressive weight training methods, we achieved superior performance across benchmark datasets, including FF++, WLDR, Celeb-DF, and DFDC, outperforming existing state-of-the-art approaches. We conducted intra-class validation across different classes of FF++ datasets to evaluate the generalization capabilities of the proposed network. There is always a need to provide an effective way to mitigate the risks associated with media and dealings to ensure security in digital content with greater trust. Our study significantly improves DDs in consumer imaging technologies.

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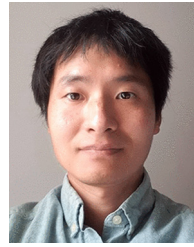


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